



$$PQ = \begin{pmatrix} 1 & 2 & \dots & n \\ p_1 & p_2 & \dots & p_n \end{pmatrix} \begin{pmatrix} 1 & 2 & \dots & n \\ q_1 & q_2 & \dots & q_n \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & \dots & n \\ p_{q_1} & p_{q_2} & \dots & p_{q_n} \end{pmatrix}$$

$$\begin{matrix} (12) \\ P \end{matrix} \begin{matrix} (13) \\ Q \end{matrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 & 2 \end{pmatrix}$$

## Multiplication table of $S_3$

| $S_3$   | $e$     | $(132)$ | $(123)$ | $(23)$  | $(13)$  | $(12)$  |
|---------|---------|---------|---------|---------|---------|---------|
| $e$     | $e$     | $(132)$ | $(123)$ | $(23)$  | $(13)$  | $(12)$  |
| $(132)$ | $(132)$ | $(123)$ | $e$     | $(12)$  | $(13)$  | $(23)$  |
| $(123)$ | $(123)$ | $e$     | $(132)$ | $(13)$  | $(23)$  | $(12)$  |
| $(23)$  | $(23)$  | $(12)$  | $(13)$  | $e$     | $(123)$ | $(132)$ |
| $(13)$  | $(13)$  | $(23)$  | $(12)$  | $(132)$ | $e$     | $(123)$ |
| $(12)$  | $(12)$  | $(13)$  | $(23)$  | $(123)$ | $(132)$ | $e$     |

$$(23)(132) \rightarrow \begin{matrix} 1 \rightarrow 3 \rightarrow 2 \\ 2 \rightarrow 1 \rightarrow 1 \\ 3 \rightarrow 2 \rightarrow 3 \end{matrix}$$

## The Dihedral group

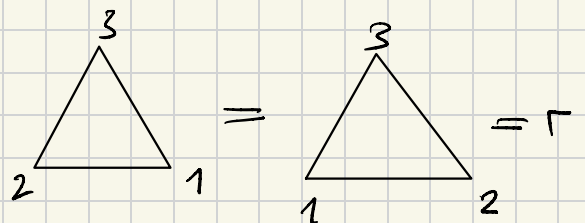
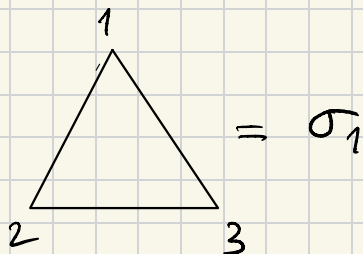
$$D_n \longrightarrow \begin{cases} \bullet \text{ Rotations } \text{angle } \frac{2\pi}{n} \\ \bullet \text{ Reflection} \end{cases}$$

Presentation  $\hookrightarrow |D_n| = 2n$

$$D_n = \langle r, \sigma \mid r^n = e, \sigma^2 = e, \sigma r \sigma = r^{-1} \rangle$$

$$D_3 = \{ \underbrace{e, r, r^{-1}}_{\text{rotations}}, \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{reflections}} \}$$

$$(\sigma_1 \sigma_2)$$



## Multiplication table of $D_3$

| $D_3$      | $e$        | $r$        | $r^{-1}$   | $\sigma_1$ | $\sigma_2$ | $\sigma_3$ |
|------------|------------|------------|------------|------------|------------|------------|
| $e$        | $e$        | $r$        | $r^{-1}$   | $\sigma_1$ | $\sigma_2$ | $\sigma_3$ |
| $r$        | $r$        | $r^{-1}$   | $e$        | $\sigma_3$ | $\sigma_1$ | $\sigma_2$ |
| $r^{-1}$   | $r^{-1}$   | $e$        | $r$        | $\sigma_2$ | $\sigma_3$ | $\sigma_1$ |
| $\sigma_1$ | $\sigma_1$ | $\sigma_2$ | $\sigma_3$ | $e$        | $r$        | $r^{-1}$   |
| $\sigma_2$ | $\sigma_2$ | $\sigma_3$ | $\sigma_1$ | $r^{-1}$   | $e$        | $r$        |
| $\sigma_3$ | $\sigma_3$ | $\sigma_1$ | $\sigma_2$ | $r$        | $r^{-1}$   | $e$        |

## Generators and rank of a group

### Def. of Generator

A subset  $S \subseteq G$  is called a set of generators of  $G$  if every element of  $G$  can be written as a finite product of elements of  $S$  and their inverses

$$G = \langle S \rangle$$

Ex / Cyclic group  $C_3$

$$C_3 = \{ e, g, g^2 \} \quad g^3 = e$$

$$\langle g \rangle = C_3$$

$$\text{rank}(C_1) = 1$$

Ex Dihedral  $D_3 = \{e, r, r^{-1}, \sigma_1, \sigma_2, \sigma_3\}$

$$\sigma = \sigma_1$$

$$\sigma_2 = r \sigma_1 = r \sigma$$

$$\sigma_3 = r^2 \sigma = r^{-1} \sigma$$

$$D_3 = \langle r, \sigma \rangle$$

$$\text{rank}(D_3) = 2$$

## Group morphism

### Homomorphism

$G$  (comp. law  $\circ$ )

$H$  (comp. law  $\diamond$ )

A homomorphism is a mapping

$$f: G \rightarrow H$$

$$g \rightarrow h = f(g)$$

Such that  $g_1 \circ g_2 = g_3 \in G$

then  $f(g_1) \diamond f(g_2) = f(g_3) \in H$

$$D_3 = \{e, r, r^{-1}, \sigma_1, \sigma_2, \sigma_3\}$$

$$C_2 = \{e, c\}, c^2 = e$$

$$f: D_3 \rightarrow C_2$$

$$e, r, r^{-1} \rightarrow e$$

$$\sigma_1, \sigma_2, \sigma_3 \rightarrow c$$

$$\textcircled{\star} \sigma_1 \circ \sigma_2 = r$$

$$\Rightarrow f(\sigma_1 \circ \sigma_2) = f(r) = e(c_2)$$

$$f(\sigma_1) \diamond f(\sigma_2) = c \diamond c = e(c_2)$$

✓

$$\textcircled{\star} \quad \Gamma \circ \sigma_1 = \sigma_2$$

$$f(\Gamma \circ \sigma_1) = f(\sigma_2) = e$$

$$f(\Gamma) \diamond f(\sigma_1) = e \diamond e = e \quad \checkmark$$

Counter example

$$\begin{cases} e & \sigma_3 & \Gamma^{-1} & \rightarrow & e \\ \sigma_1 & \sigma_2 & \Gamma & \rightarrow & e \end{cases}$$

$$\sigma_3 \circ \Gamma^{-1} = \sigma_2$$

$$f(\sigma_3 \circ \Gamma^{-1}) = f(\sigma_2) = e$$

$$f(\sigma_3) \diamond f(\Gamma^{-1}) = e \diamond e = e \quad \times$$

Kernel

Kernel of a homomorphism  $f$  measures "how much" of the structure of  $G$  collapses on the identity of the map  $f$ .

$$\ker(D_3) = \{e, \Gamma, \Gamma^{-1}\}$$

Isomorphism

$\hookrightarrow$  bijective homomorphism !

$$G \cong H$$

For finite group isomorphisms  $\equiv$  identical multiplication table

$$C_4 = \{e, a, a^2, a^3\}, \quad a^4 = e$$

$$G_1 = \{1, i, -1, -i\}$$

$$\Rightarrow C_4 \cong G_1 \cong G_3$$

$$G_3 = \{1, i\sigma_2, -1, -i\sigma_2\}, \quad \sigma_2 = \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix}$$

## Endomorphism

↳ homomorphism of a group  $G$  on itself  $f: G \rightarrow G$

Ex:  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 2x$   
(with addition)

## Automorphism

↳ bijective endomorphism

The set of all automorphism of a group  $G$  is denoted

$$\text{Aut}(G) = \{ f: G \rightarrow G \mid f \text{ bijective and } f(gh) = f(g)f(h) \}$$

Ex/ group  $(\mathbb{Z}, +) \rightarrow$  two automorphism  $f(n) = n$   
 $f(n) = -n$

Therefore  $\text{Aut}(\mathbb{Z}, +) \simeq C_2$

## Cayley's Theorem

Every finite group of order  $n$  is isomorphic to a subgroup of the symmetric group  $S_n$